Best Practices for Biomedical Research Data Management

Module 1: Introduction and Overview Research Data Management

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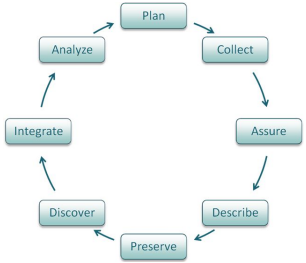


This video will provide an overview of “Research Data Management” and the many areas involved. It is intended to expand on the concepts in the introductory slides and give some examples.

BY NC SA

What is Research Data?

"the recorded factual material commonly accepted in the scientific community as necessary to validate research findings, but not any of the following:

preliminary analyses, drafts of scientific papers, plans for future research, peer reviews, or communications with colleagues;

trade secrets, commercial information, materials necessary to be held confidential by a researcher until they are published, or similar information which is protected under law;

personnel and medical information and similar information the disclosure of which would constitute a clearly unwarranted invasion of personal privacy, such as information that could be used to identify a particular person in a research study."

Office of Management and Budget Circular A-110 (http://www.whitehouse.gov/omb/circulars_a110#36).

What is Research Data Management

Data Management refers to the storage, access and preservation of data produced from a given investigation. Data management practices cover the entire lifecycle of the data, from planning the investigation to conducting it, and from backing up data as it is created and used to long term preservation of data deliverables after the research investigation has concluded.

Casrai Dictionary: http://dictionary.casrai.org/Research_data_management

Why Manage Research Data?

Data management is a core part of **Responsible Conduct of Research**

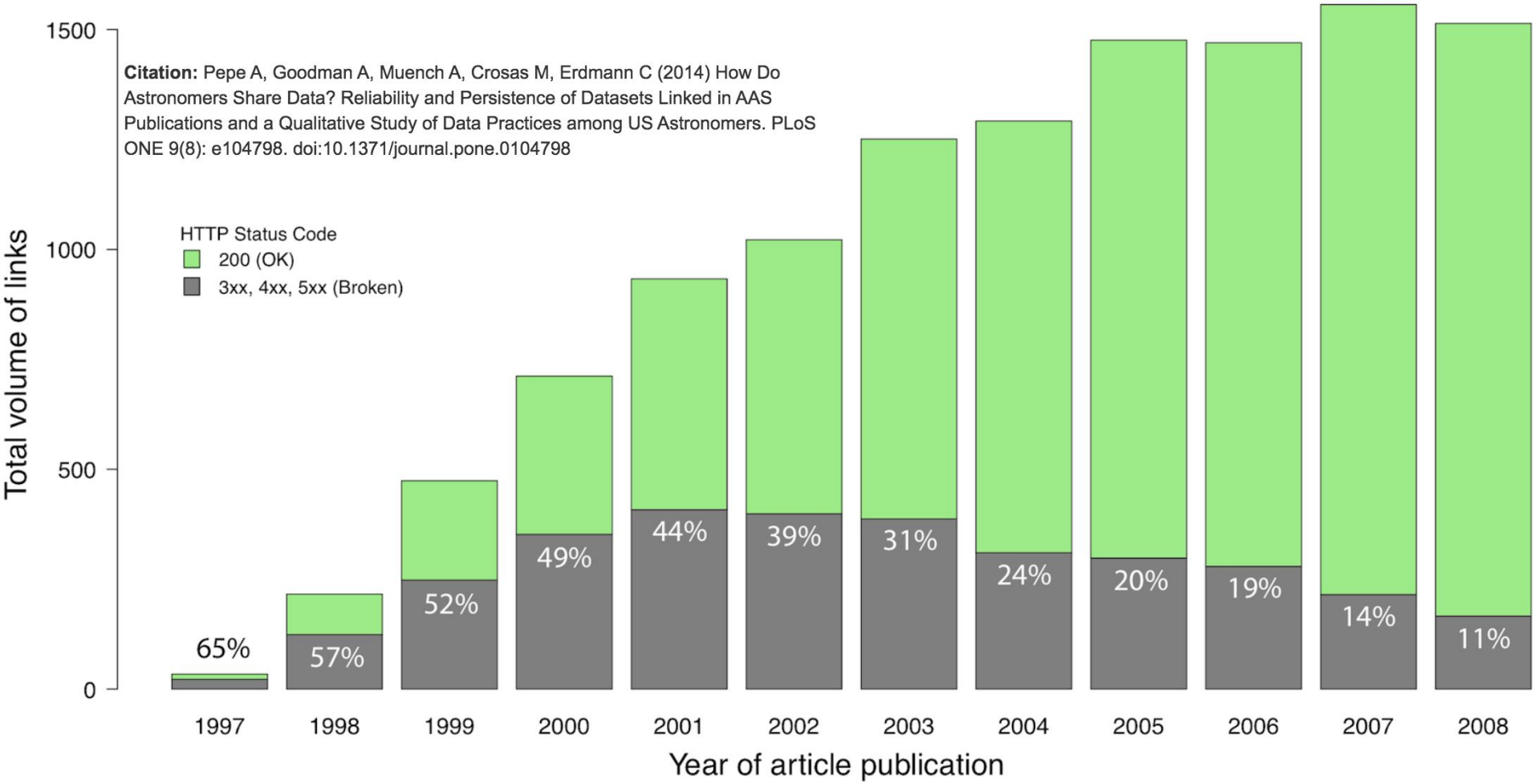
Data sets have **value**; at least equivalent in value to the articles published based upon them

Data sets can be shared with and reused and repurposed by other researchers to **further discovery**

Institutions and funders want data to be **managed and retained** in the case of an audit or investigation

Federal funders want publicly-funded research results, including data, to be **disseminated** and **publicly available**

Data sets are **at risk**


Citation: Pepe A, Goodman A, Muench A, Crosas M, Erdmann C (2014) How Do Astronomers Share Data? Reliability and Persistence of Datasets Linked in AAS Publications and a Qualitative Study of Data Practices among US Astronomers. PLoS ONE 9(8): e104798. doi:10.1371/journal.pone.0104798
HTTP Status Code
200 (OK)
3xx, 4xx, 5xx (Broken)
Total volume of links
1500
1000
500
0
65%
57%
52%
49%
44%
39%
31%
24%
20%
19%
14%
11%
1997
1998
1999
2000
2001
2002
2003
2004
2005
2006
2007
2008
Year of article publication

RDM: Managing the Recipe for Reproducibility

Data: appraising data underlying published results

Metadata and documentation: contextual information and details about the data sets needed for discovery, access, use, and reuse

Code and scripts associate with analyses and results

The Lifecycle of Research Data

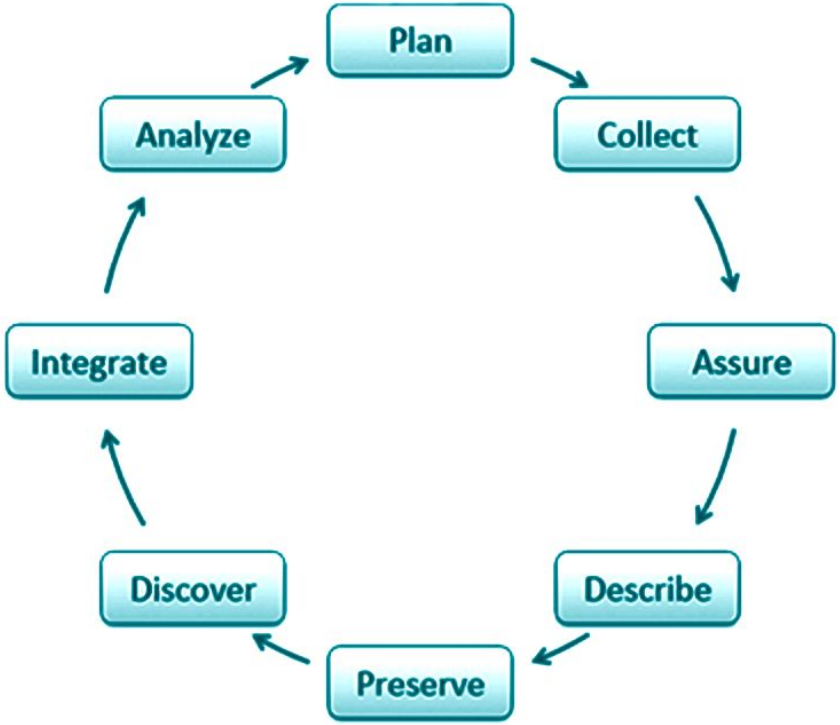


Image credit: Dataone Leadership Team https://www.dataone.org/data-life-cycle

Plan: Before Data Collection Begins...

Create a standard for **file naming conventions** among team members

Create a standard way for **structuring folder directories**

Utilize any **community standards** for naming and **describing data** elements

Utilize **reporting standards** and **minimum information standards** to be sure to record sufficient contextual information about experiments

Choose **file formats** optimal for **storage and archiving** and ones for optimal for **dissemination, access, use and reuse**

Explore **data sharing repositories** early so that data can be described and formatted in line with the repository's guidelines for deposit from the beginning

Describe: For Discovery, Access, Use and Reuse

Controlled vocabulary for naming data elements and standards for defining values

Data dictionaries

Readme files

Codebooks

Minimum Information and reporting standards

Preserve Access: Repositories and Data Centers

Disciplinary repositories

Repositories built around methods

Federally-funded repositories and data centers

Institutional repositories

Open repositories

Data enclave options for sensitive data sets

Discover: Publishing and Citing Data

Obtaining digital object identifiers (DOIs) for data sets

Utilizing other identifiers: accession numbers (NCBI), clinical trial registry numbers (NCT)

Citing data sets and their location in publications and presentations along with funding source, as well as in grant reports and future grant proposals

Assigning a license defining terms of use

Ensuring sensitive data sets meet IRB standards for consent and de-identification; deposit after appropriate embargoing for publication, intellectual property or security

Citation: Chrastil ER, Warren WH (2014) From Cognitive Maps to Cognitive Graphs. PLoS ONE 9(11): e112544. doi:10.1371/journal.pone.0112544

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Data Availability: The authors confirm that all data underlying the findings are fully available without restriction. Data are available from the Brown University Digital Repository (https://repository.library.brown.edu/studio/item/bdr:384192/) using the DOI 10.7301/Z0KW5CXV.

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Data Policies

Data ownership and intellectual property

Data management and stewardship responsibilities

Public access, data sharing, and dissemination

Retention

U.S.Department of Health & Human Services

NIH Genomic Data Sharing (GDS)

http://datasharing.sparcopen.org

SPARC

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Browse Data Sharing Requirements by Federal Agency

This community resource for tracking, comparing, and understanding both current and future U.S. federal funder research data sharing policies is a joint project of SPARC & Johns Hopkins University Libraries. Click the icons below to select up to three agencies to view or compare. Click here to download the full data set.

Search for an agency...

AHRQ

ASPR

CDC
CENTERS FOR DISEASE CONTROL AND PREVENTION

DEPARTMENT OF DEFENSE
UNITED STATES OF AMERICA

DEPARTMENT OF EDUCATION
UNITED STATES OF AMERICA

The NEW ENGLAND JOURNAL of MEDICINE

CORRESPONDENCE

RETRACTION

Retraction: CPAP for the Metabolic Syndrome in Patients with Obstructive Sleep Apnea. N Engl J Med 2011;365:2277-86

N Engl J Med 2013; 369:1770 | October 31, 2013 | DOI: 10.1056/NEJMc1313105

To the Editor:

As the authors of the article entitled "CPAP for the Metabolic Syndrome in Patients with Obstructive Sleep Apnea"[1] published in the *Journal* on December 15, 2011, we regret to report that transcription errors occurred in the Supplementary Appendix, available with the full text of the article at NEJM.org. There were multiple errors in the table on pages 18 and 19 of the Supplementary Appendix concerning data on the accumulation of abdominal fat as assessed with the use of computed tomography and on carotid intima–media thickness as assessed with the use of ultrasonography. These errors, in turn, changed some values in Table 4 of the article. Although these changes do not alter the conclusions of the article, the primary data could not be located to verify corrections made from secondary tables. Accordingly, we have no way of confirming the correct data and, with regret, wish to retract the article.

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